Graham World Cup Hub Model: Algorithm and Variable Reference

Expected-goals forecasting for FIFA World Cup 2026 fixtures

Match Predictor · Technical documentation · Generated 28 June 2026

ABSTRACT

This document specifies the Graham World Cup hub model used exclusively for FIFA World Cup match prediction. The pipeline combines decay-weighted international process rates, shot-quality profiles, a six-component strength index (ΔS), in-tournament Opta composites, venue and motivation adjustments, optional lineup-aware blending, and Dixon–Coles score correlation to produce expected goals and 1X2 outcomes. Coefficients are bounded and updated post-match via calibration when sufficient finished fixtures exist.

1. Pipeline Overview

Historical international matches + WC 2026 results → process attack/defense rates
Opta WC player/match data → tournament composites + lineup form
→ ΔS strength index → baseline home/away xG
→ momentum + finishing regression + set-piece bump → venue (altitude, rest, jet lag, host, motivation)
→ optional Model XI lineup blend → Dixon–Coles score grid (ρ)
→ 1X2, scoreline, over/under 2.5

Entry point: `runGrahamWorldCupPredict` → `resolveGrahamExpectedGoals` → venue/motivation → optional lineup → `outcomesFromGuardedGrid`. The club prediction engine is *not* used for World Cup matches.

Calibrated constants load from Supabase `world_cup_calibration_config` (latest saved version: `wc-graham-v1.12-md12`). Defaults below apply when nothing is saved.

2. Process Attack and Defense Rates

Function: `computeGrahamProcessRatesFromMatches` → optionally `applyWcFormToProcessRates`

2.1 Match weights and rolling averages

For each past match m for team T:

$$w_m = \mathrm{decay}(\mathrm{date}_m) \times \mathrm{tier}(\mathrm{competition}_m) \times c_{\mathrm{opp}}$$

$$\overline{xGF} = \frac{\sum_m xGF_m \cdot w_m}{\sum_m w_m}, \qquad \overline{xGA} = \frac{\sum_m xGA_m \cdot w_m}{\sum_m w_m}$$

2.2 Raw rates and shrinkage

$$\text{attack}_{\text{raw}} = \text{clamp}\left(\frac{\overline{xGF}}{\mu},\ 0.4,\ 2.5\right), \qquad \text{defense}_{\text{raw}} = \text{clamp}\left(\frac{\overline{xGA}}{\mu},\ 0.4,\ 2.5\right)$$

$$\text{attack} = 1 + (\text{attack}_{\text{raw}} - 1) \cdot \frac{w}{w+K}, \qquad \text{defense} = 1 + (\text{defense}_{\text{raw}} - 1) \cdot \frac{w}{w+K}$$

2.3 WC tournament nudge

Applied only when the team has at least one WC 2026 match ingested:

$$\text{attack}' = 1 + (\text{attack} \cdot n_{\text{atk}} - 1) \cdot \omega_{\text{atk}} + (\text{attack} - 1) \cdot (1 - \omega_{\text{atk}})$$

$$\text{defense}' = 1 + \left(\frac{\text{defense}}{n_{\text{def}}} - 1\right) \cdot \omega_{\text{def}} + (\text{defense} - 1) \cdot (1 - \omega_{\text{def}})$$

Symbol	Default	Meaning
μ	**1.25** (`muXg`)	Average team xG per game anchor
Decay φ	**0.00048**/day	Exponential time decay (~4-year half-life)
Tier weights	Friendly **0.32**, competitive **1.0**, WC finals **1.0**	Friendlies count less
c_{opp}	Confederation modifier	Opponent region strength adjustment
K	**5**	Shrinkage toward 1.0 when few games
Rate clamps	**0.4 – 2.5**	Outlier bounds
SOT→xG fallback	**0.32**	Proxy when xG missing
Shots→xG fallback	**0.10**	Weaker proxy
$\omega_{\text{atk}}, \omega_{\text{def}}$	**0.35** each	WC in-tournament form blend weight
$n_{\text{atk}}, n_{\text{def}}$	**1 ± 8%** max	From Opta composites (Section 6)

> **Plain-language summary.** Count how many chances teams created and allowed in past games (recent games matter more). Then give a small extra push based on how they have played in this World Cup.

3. Shot Quality Profiles (SCI / SSI)

Function: `computeShotProfileFromMatches` → `applyShotProfilesForFixture`

$$SCI = \frac{xGF}{\text{shots}}, \qquad SSI = \frac{xGA}{\text{oppShots}}$$

$$\text{homeAttack} \leftarrow \text{homeAttack} \cdot \big(1 + 0.08 \cdot (SCI_{\text{home}} - SCI_{\text{league}})\big)$$

$$\text{homeDefense} \leftarrow \text{homeDefense} \cdot \big(1 + 0.08 \cdot (SSI_{\text{league}} - SSI_{\text{home}})\big)$$

(Symmetric adjustments for the away team.)

Coefficient	Value
`GRAHAM_SCI_WEIGHT`	**0.08**
`GRAHAM_SSI_WEIGHT`	**0.08**
League SCI/SSI	Average of both teams in the fixture

> **Plain-language summary.** Not just how many shots — how good each shot tends to be. Teams that need many shots for one chance get a small downgrade.

4. Recent Form xG (feeds ΔS)

$$\text{recentForm}_{\text{home}} = \mu \cdot \text{homeAttack} \cdot \text{awayDefense}$$

$$\text{recentForm}_{\text{away}} = \mu \cdot \text{awayAttack} \cdot \text{homeDefense}$$

$$\Delta\,\text{recentForm} = \text{recentForm}_{\text{home}} - \text{recentForm}_{\text{away}}$$

> **Plain-language summary.** If these two teams played right now based on recent chance creation, home would expect one xG level and away another. This difference feeds the strength index.

5. Core Strength Index and Baseline xG

Function: `resolveGrahamExpectedGoals`

5.1 Component deltas

Component	Formula	Scale into ΔS
xG-Elo	$\text{homeXgElo} - \text{awayXgElo}$	$\times\ w_{\text{xgElo}}$
Talent	$(\text{homeTalent} - \text{awayTalent}) \times 400$	$\times\ w_{\text{talent}}$
Tournament (WCTR)	$\text{homeWctr} - \text{awayWctr}$	$\times\ w_{\text{tournament}}$
Recent form	$\Delta\,\text{recentForm} \times 100$	$\times\ w_{\text{recent}}$
FIFA	$\text{homeFifaPts} - \text{awayFifaPts}$	$\times\ w_{\text{fifa}}$
Momentum	$\text{momentum} \times 120$	$\times\ w_{\text{momentum}}$

5.2 Strength index and exponential xG mapping

$$\Delta S = \sum_i w_i \cdot \Delta_i$$

$$\lambda_{\text{home,base}} = \text{clamp}\big(\mu \cdot e^{c \cdot \Delta S},\ 0.1,\ 5.0\big)$$

$$\lambda_{\text{away,base}} = \text{clamp}\big(\mu \cdot e^{-c \cdot \Delta S},\ 0.1,\ 5.0\big)$$

5.3 Default ΔS weights (sum = 1)

Weight	Default
w_{xgElo}	**40%**
w_{talent}	**20%**
$w_{\text{tournament}}$	**15%**
w_{recent}	**10%**
w_{fifa}	**10%**
w_{momentum}	**5%**

Coefficient	Default	Role
c (`strengthExponent`)	**0.00305**	Steepness of strength → xG mapping
μ (`muXg`)	**1.25**	Baseline goals per team
xG floor/cap	**0.1 – 5.0**	Hard bounds on expected goals

5.4 Sub-model inputs

- **xG-Elo** (`computeXgEloFromMatches`): Elo-style rating updated by $(\text{actual xG diff} - \text{expected diff}) \times K$, with $K = 0.32\times$ match tier. Initialized from FIFA points.
- **WCTR** (`computeWctrFromMatches`): Same machinery on tournament-weighted matches (WC/Euro knockouts up to **1.25×**).
- **Talent** (`resolveSquadTalentSnapshot`): squad value = **60%** Transfermarkt + **40%** Scoutlyst; $\text{talentRating} = \log(\text{squadValue}/\text{medianWC})$.
- **Momentum** (`computeGrahamMomentumIndex`): $\text{momentum} = (\text{homeFormScore} - \text{awayFormScore}) \cdot 0.55 + \text{h2h} \cdot 0.10$, clamped to **±0.65**, then applied as $\lambda_{\text{home}} \leftarrow \lambda_{\text{home}} \cdot e^{0.015 \cdot \text{momentum}}$ and $\lambda_{\text{away}} \leftarrow \lambda_{\text{away}} \cdot e^{-0.015 \cdot 0.92 \cdot \text{momentum}}$.

> **Plain-language summary.** Build one "who is stronger?" score from six signals, then turn that into home and away expected goals.

6. WC 2026 In-Tournament Form (Opta Composites)

Source: Opta player stats → `world_cup_team_match_aggregates` → `loadWcInTournamentFormNudges`

6.1 Five composites per team per match

Composite	Built from (raw Opta)	Role
chanceIndex	Player xG×**1.2** + shot created×**0.18** + in-box×**0.06**, minutes-weighted, winsorized ±**2.5**	Attack signal
defensiveSolidity	Tackles×**0.35** + interceptions×**0.4** + blocks×**0.25** + SOT against×**0.15**	Defense signal
finishingDelta	goals $-$ team_xG, winsorized ±**2.5**	Over/under-performance
territoryIndex	Possession×**0.45** + final-third×**0.35** + penalty-area×**0.2**	Pitch control
gkSaveIndex	saves×**0.5** − goals×**0.3** + pen saves×**0.8**	Goalkeeper quality
disciplineLoad	yellows×**0.25** + reds×**1.5** per player	Cards / availability
opponentStrength	FIFA/1400, clamped **0.65–1.45**	Opponent adjustment

6.2 Opponent-adjusted tournament average

$$\bar{x} = \frac{\sum_m (x_m / s_m^{\text{opp}}) \cdot \left(1/\max(0.7, s_m^{\text{opp}})\right)}{\sum_m 1/\max(0.7, s_m^{\text{opp}})}$$

6.3 Nudges and finishing regression

$$\text{attackSignal} = \text{chanceIndex} \cdot 0.12 + (\text{territory} - 0.5) \cdot 0.25$$

$$n_{\text{atk}} = \text{clamp}(1 + \text{attackSignal},\ 0.92,\ 1.08)$$

$$n_{\text{def}} = \text{clamp}(1 + \text{defensiveSolidity} \cdot 0.10,\ 0.92,\ 1.08)$$

$$\text{finishingRegression} = \text{clamp}(-\text{finishingDelta} \cdot 0.15,\ -0.06,\ 0.06)$$

$$\lambda_{\text{home}} \leftarrow \lambda_{\text{home}} \cdot (1 + \text{finishingRegression}_{\text{home}})$$

Calibrated weight	**Default**
`wcAttackFormWeight` (ω_{atk})	**0.35**
`wcDefenseFormWeight` (ω_{def})	**0.35**
`wcFinishingRegressionWeight`	**0.15**

Squad market-value talent weights are *not* modified by Opta (by design).

> **Plain-language summary.** After each WC game we summarize how teams played in five scores. Strong chance creation or defending nudges rates slightly; lucky finishing is gently regressed toward xG.

7. Set-Piece Bump

If away team set-piece goal share ≥ **40%** (`setPieceRateThreshold`):

$$\lambda_{\text{away}} \leftarrow \lambda_{\text{away}} + 0.15$$

Rates come from Opta article ingests per team.

8. Venue, Travel, and Group Motivation

Function: `runGrahamWorldCupPredict` (after baseline xG)

$$\lambda_{\text{home}} \leftarrow \lambda_{\text{home}} \cdot \gamma_{\text{alt}} \cdot \delta_{\text{home}} \cdot \sigma_{\text{home}} \cdot \text{hostBoost}$$

$$\lambda_{\text{away}} \leftarrow \lambda_{\text{away}} \cdot \gamma_{\text{alt}} \cdot \delta_{\text{away}} \cdot \sigma_{\text{away}}$$

Factor	Rule	Typical value
Altitude γ_{alt}	≤1500 m: **1.0**; else **0.96**	Thin air → fewer goals
Rest δ_{rest}	72 h+ → **1.0**; 0 h → **0.85**; linear between	Fatigue penalty
Jet lag θ	Eastward: **−1.5%**/hour TZ; westward: **−0.5%**	Travel fatigue
δ_{final}	$\text{clamp}(\delta_{\text{rest}} \times \text{jetLag},\ 0.82,\ 1.0)$	Combined travel
Host boost	Mexico City ×**1.05**; USA/Canada hosts ×**1.04**	Home advantage
σ (motivation)	Group winner clinched → **0.92–0.95**	Rotation risk
ρ offset	Mutual advancement on draw → **−0.12**	Strategic draws

> **Plain-language summary.** Adjust goals for tiredness, travel, altitude, home-country advantage, and whether both teams want to attack.

9. Model XI Lineup Impact

Function: `computeWcLineupPlayerXgImpact` → `applyLineupImpactToHubPrediction`

Per player in projected/matched XI:

$$\text{attackContrib} = (\mu \cdot 0.12 + \text{chanceIndexPer90} \cdot 0.85 + \text{optaNorm} \cdot 0.35) \cdot \text{roleBoost} \cdot \text{availabilit}$$

$$\lambda_{\text{lineup}} = 0.95 \cdot \sum \text{attackContrib}$$

$$\lambda_{\text{blended}} = (1-\omega) \cdot \lambda_{\text{base}} + \omega \cdot \lambda_{\text{lineup}}, \qquad \omega = 0.35 \text{ (or 0.175 if } < 7 \text{ players matched)}$$

$$\lambda_{\text{home,final}} = \lambda_{\text{base,home}} \cdot \alpha_{\text{home}} \cdot \beta_{\text{away}}$$

Defense multipliers cross-apply: weak home defense boosts away xG.

Coefficient	Value
`wcLineupAttackBlend`	**0.35**
`wcLineupDefenseBlend`	**0.35**
Attack mult clamp	**0.78 – 1.14**
Defense mult clamp	**0.86 – 1.18**
`availabilityFactor`	**1 − fatigue×0.12 − cards×0.08**, min **0.55**
Role boosts	F ×**1.08**, M ×**1.02**, D/G ×**0.95**

> **Plain-language summary.** Who is starting and how they have played in the World Cup nudges expected goals up or down.

10. Score Correlation and Match Outcomes

Function: `outcomesFromGuardedGrid` (Dixon–Coles Poisson)

$$P(h,a) \propto \text{Poisson}(h;\, \lambda_{\text{home}}) \cdot \text{Poisson}(a;\, \lambda_{\text{away}}) \cdot \tau_{h,a}(\rho)$$

$$\rho_{\text{base}} = f(|\lambda_{\text{home}} - \lambda_{\text{away}}|,\ \lambda_{\text{home}} + \lambda_{\text{away}},\ |\Delta\text{FIFA}|) + \rho_{\text{motivation}}$$

10.1 ρ rules

Condition	ρ
Effective gap ≥ 1.25	**+0.10**
Gap ≥ 0.75	**+0.06**
Gap ≥ 0.28	**+0.04**
Total xG ≥ 3.1	**−0.08**
Total xG ≥ 2.4	**−0.11**
Else	**−0.14**

ρ attenuates toward 0 as the xG gap widens. Low-event WC boost: if both teams' `avgChanceIndex < 1.2`, then $\rho \leftarrow \rho + 0.025$.

Outputs: home win %, draw %, away win %, modal scoreline, over/under 2.5 %.

> **Plain-language summary.** Build every plausible scoreline and add extra draw probability when the game looks tight or both teams might settle for a point.

11. Post-Match Calibration

Function: `wc-calibrate-graham` after ≥2 evaluated matches. Tunable within **±10%** of defaults:

Constant	Default	What it moves
`muXg` (μ)	1.25	Overall goal level
`strengthExponent` (c)	0.00305	Strength → xG steepness
`deltaWeights` (6)	see Section 5.3	Strength signal mix
`wcAttackFormWeight`	0.35	WC form on attack rate
`wcDefenseFormWeight`	0.35	WC form on defense rate
`wcFinishingRegressionWeight`	0.15	Luck correction
`wcLineupAttackBlend`	0.35	XI attack blend
`wcLineupDefenseBlend`	0.35	XI defense blend
`wcLowEventRhoBoost`	0.025	Extra draw mass (low chance)
`setPieceXgBump`	0.15	Set-piece team bump
`setPieceRateThreshold`	0.40	When bump applies

Individual raw Opta stat weights are *not* calibrated (composites only).

12. Variable Checklist

A. Historical match inputs

`home_xg, away_xg, home_shots, away_shots, home_sot, away_sot, home_goals, away_goals, competition, date, opponent confederation`

B. Strength / structural

FIFA ranking points, xG-Elo, WCTR, squad market value (TM/Scoutlyst), median squad value, momentum index, H2H xG edge

C. WC 2026 Opta-derived

Player: xG, shot created, in-box attempts, tackles, interceptions, blocks, saves, goals, cards, minutes, Opta Points. Team composites: chanceIndex, defensiveSolidity, finishingDelta, territoryIndex, gkSaveIndex, disciplineLoad, opponentStrength. Player tournament form: avgOptaPoints, chanceIndexPer90, defensiveActionsPer90, gkSaveIndex, availabilityFactor, wasLastStarter.

D. Match context

Venue altitude, rest hours, timezone travel, host nation, group standings (σ, ρ offset), set-piece rate

E. Outputs

λ_{home}, λ_{away}, 1X2 probabilities, predicted score, O/U 2.5

13. Analyst Notes

1. **Two layers of form:** long international history (decayed xG rates) plus WC-only Opta composites (bounded ±8%).
2. **Talent is separate:** squad value enters ΔS at 20%; Opta does not change talent.
3. **Lineup matters:** hub predictions use projected last-match starters, not neutral team xG.
4. **Draws are explicit:** Dixon–Coles ρ, group-motivation offsets, and low-event boost.
5. **Coefficients evolve slowly:** `wc:postmatch` → evaluate → calibrate saves new constants when enough matches exist.